



Segmentation of PLS path models by iterative reweighted regressions[☆]



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ARTICLE INFO

Article history:

Received 1 September 2015

Received in revised form 1 November 2015

Accepted 1 March 2016

Available online 26 April 2016

Keywords:

Partial least squares

PLS

PLS-IRRS

Reweighted regressions

Segmentation

Genetic algorithms

Fuzzy set qualitative comparative analysis

fsQCA

ABSTRACT

Uncovering unobserved heterogeneity is a requirement to obtain valid results when using structural equation modeling (SEM). Conventional segmentation methods usually fail in an SEM context because they account for the indicator data, but not for the latent variables and their relationships in the structural model. This research introduces a new segmentation approach to variance-based SEM using partial least squares path modeling (PLS). The iterative reweighted regressions segmentation method for PLS (PLS-IRRS) effectively identifies and treats unobserved heterogeneity in data sets. Compared to existing alternatives, PLS-IRRS is multiple times faster while delivering results of the same quality. Researchers should therefore routinely use PLS-IRRS to address the critical issue of unobserved heterogeneity in PLS.

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1. Introduction

Partial least squares path modeling (PLS; Chin, 1998; Dijkstra, 2010; Lohmöller, 1989; Wold, 1982) is a variance-based method to estimate composite-based path models (Henseler, Ringle, & Sarstedt, 2016a; Henseler et al., 2014). Being a prediction-oriented approach to SEM, the PLS method is particularly useful when the study focuses on analyzing a certain target construct's key sources of explanation. PLS enjoys a rapidly increasing usage in various business disciplines such as accounting (Lee, Petter, Fayard, & Robinson, 2011), family business (Sarstedt, Ringle, Smith, Reams, & Hair, 2014a), management information systems (Ringle, Sarstedt, & Straub, 2012), marketing (Hair, Sarstedt, Ringle, & Mena, 2012c), operations management (Peng & Lai, 2012), strategic management (Hair, Sarstedt, Pieper, & Ringle, 2012b), and tourism (do Valle & Assaker, 2015)—as numerous review studies show. Even though the popularity of PLS continues to increase, researcher often call for more rigor when applying the method (e.g., Hair, Ringle, &

Sarstedt, 2012a, 2013; Rigdon et al., 2014; Sarstedt, Ringle, & Hair, 2014c). One recommendation recited like a mantra (e.g., Hahn, Johnson, Herrmann, & Huber, 2002; Hair, Sarstedt, Matthews, & Ringle, 2016; Hair et al., 2012c; Sarstedt, 2008a) concerns the identification of unobserved heterogeneity, which, if neglected, is a serious threat to the validity of results (Becker, Rai, Ringle, & Völckner, 2013b; Jedidi, Jagpal, & DeSarbo, 1997). In situations where unobserved heterogeneity is present, researchers cannot generalize the results from the aggregate data level analysis, but must account for differences in model relationships by forming adequate segments of observations.

Conventional segmentation methods such as k-means perform poorly in an SEM context because they ignore the path model relationships that the researchers specified prior to the analysis (Sarstedt & Ringle, 2010). For this reason, research has proposed a wide array of latent class techniques, which generalize, for example, finite mixture, typological regression or hill-climbing approaches to PLS (for an early overview, see Sarstedt, 2008b). In this stream of research, Ringle, Sarstedt, Schlittgen, and Taylor (2013) proposed the PLS genetic algorithm segmentation (PLS-GAS) method (also see Ringle, Sarstedt, & Schlittgen, 2014), which uses guided random searches to disclose an optimum solution in the complex search space that underlies segmentation tasks in PLS. Implementing a genetic algorithm in a first step, PLS-GAS investigates many constellations before stopping in a situation, which is at least close to the global optimum. In a second step, a deterministic hill-climbing approach locally improves the solution from the first step, if possible. Ringle et al.'s (2013) and Ringle et al.'s (2014)

[☆] The authors thank Joseph F. Hair, Jr. (Kennesaw State University) and Edward E. Rigdon (Georgia State University) for their helpful remarks on prior versions of this paper. The authors alone are responsible for all limitations and errors that may relate to the study and the paper. This article uses the statistical software SmartPLS 3 (<http://www.smartpls.com>). Becker and Ringle acknowledge a financial interest in SmartPLS.

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simulation studies reveal that PLS-GAS outperforms the commonly used finite mixture PLS (FIMIX-PLS; Hahn et al., 2002) and response-based unit segmentation (REBUS-PLS; Esposito Vinzi, Trinchera, Squillacciotti, & Tenenhaus, 2008) latent class approaches in terms of classification accuracy and predictive power across a broad range of data and model constellations. In the light of these results, the authors call for the routine use of PLS-GAS in future research.

While commendable, a major obstacle in following this call relates to the methods run times, resulting from the great number of potential solutions considered in both stages of PLS-GAS. Table 1 illustrates PLS-GAS run times for some sample PLS-based segmentation studies from prior research. The table shows that the computations easily take more than one hour for a single PLS-GAS run. Hence, a full-scale PLS-GAS analysis (e.g., analysis of two to ten pre-specified numbers of segments; 20 PLS-GAS runs per pre-specified number of segments) may take a week or longer on a standard laptop computer. As Matthews, Sarstedt, Hair, and Ringle (2016) note, such run times pose a serious problem for PLS-GAS's application in studies, hindering its dissemination in research practice.

Addressing this limitation of PLS-GAS, this paper presents the iterative reweighted regressions segmentation method for PLS (PLS-IRRS), a new latent class approach to PLS, which offers at least the same segmentation quality as PLS-GAS but at a considerably shorter run time (Table 1). PLS-IRRS builds on ideas related to M-estimators (Schlittgen, 2011) to identify homogeneous segments of observations, which have similar residual values. The PLS-IRRS method does not rely on distributional assumptions and, thus, matches PLS's nonparametric character. Furthermore, PLS-IRRS allows uncovering heterogeneity in path models that involve both reflective and formative measurement models. Thereby, the method is generally applicable to segmentation tasks in PLS path models commonly encountered in business research studies (e.g., Fiedler & Sarstedt, 2014; Ringle, Sarstedt, & Mooi, 2010; Valette-Florence, Guizanib, & Merunka, 2011).

Results from a large-scale simulation study show that PLS-IRRS effectively and efficiently identifies unobserved heterogeneity. Compared to PLS-GAS—the best available alternative for segmentation in PLS (Ringle et al., 2013, 2014)—the PLS-IRRS method is several hundred times faster while delivering results of the same quality. This finding suggests that researchers should routinely use PLS-IRRS to address the critical issue of unobserved heterogeneity in PLS, thereby meeting the requirements brought forward by multiple author groups (Becker et al., 2013b; Esposito Vinzi et al., 2008; Hahn et al., 2002; Ringle et al., 2014; Sánchez, 2009). An application of PLS-IRRS to the American Customer Satisfaction Index (ACSI) model (Anderson & Fornell, 2000; Fornell, Johnson, Anderson, Cha, & Bryant, 1996) provides further evidence for the method's usefulness. Finally, a fuzzy set qualitative comparative analysis (fsQCA; Fiss, 2011; Woodside, 2013) allows assessing causal asymmetry in segment-specific path relationships of the ACSI model.

2. The PLS-IRRS method

PLS-IRRS builds on Schlittgen's (2011) approach to clusterwise regression, a general framework for partitioning data and simultaneously estimating segment-specific regression coefficients. The method uses reweighted least squares to compute M-estimators, originally designed to mitigate the influence of outliers on regression results by weighting observations down that have extreme values for one or more dependent variable(s). Observations with large residuals in the dependent variable(s) receive smaller weights, thereby limiting their influence for the computation of segment-specific regression coefficients—and vice versa. Instead of regarding outliers as such, PLS-IRRS uses the segment-specific weights of an observation for its segment assignment. Within a certain segment (e.g., segment 1 of 3), an observation that is similar to (different from) other observations in its segment returns a small (large) residual, which results in a higher (smaller) weight of that observation. Thereby, the method estimates weighted regression solutions for each segment instead of treating all observations equally. In a final step, the approach assigns observations to the segment where each observation has the highest weight. This assignment follows the expectation that an observation fits best into that segment where the observation has the smallest residual value. Simulation study results of regression models show that the method reliably detects segment structures across a broad range of data and model constellations, outperforming commonly used mixture regressions in the vast majority of cases (Schlittgen, 2011).

As the PLS algorithm rests on a series of regressions and linear combinations involving indicator and construct data, Schlittgen's (2011) weighted least squares approach can readily be adapted within a PLS framework. The PLS-IRRS algorithm (Fig. 1) starts with a random choice of weights w_{ik} , where i indicates an observation and $k = 1, \dots, g$ the different segments ($\sum_{k=1}^g w_{ik} = 1 \quad \forall i = 1, \dots, n$). Next, the method determines the segment-specific PLS solutions accounting for the weights ($w_{1k}, \dots, w_{nk}$)' of the g segment vectors in the PLS algorithm, by multiplying the data matrix with the square root of the weights.

Building on these results, in the next step, PLS-IRRS computes new weights as the inverse of the sum of structural model residuals (i.e., r_{ik}) of all the regressions. More precisely, by using normed reciprocal values $1/r_{ik}^2$ as new weights w_{ik} , an observation i gets a higher (lower) weight in segments where the sum of its squared residuals is small (large). Using these new weights as input, PLS-IRRS updates the segment-specific solutions and, again, determines new weights. The algorithm terminates when the path coefficients and outer weights estimates stabilize (i.e., difference of estimated coefficients between two iterations reaches a value that is smaller than a pre-defined level Δ). After convergence follows the comparison of the segment-specific weights per observation. PLS-IRRS assigns observations to the segment where each observation has the highest weight and estimates segment-specific PLS path models based on this final assignment.

Table 1
PLS-GAS and PLS-IRRS run time in segmentation studies.

Reference	Model	Sample size	Number of segments	Approximate run times in seconds	
				PLS-GAS	PLS-IRRS
Ringle et al. (2010)	American Customer Satisfaction Index	10417	2	6600	<5
Wilden and Gudergan (2015)	Dynamic and operational capabilities	228	2	2600	<5
Wamba and Trinchera (2014)	Technology acceptance	2556	3	5200	<5
Sarstedt and Ringle (2010)	Corporate reputation	344	2	2600	<5
Valette-Florence et al. (2011)	Drivers of brand equity	538	3	4000	<5

Notes: Run times on a standard laptop (Intel Core i7 with 2.7 GHz, 8 GB RAM and Windows 7 64 bit); PLS-GAS settings, first stage: 100 individuals, 1000 generations; PLS-GAS setting, second stage: max. number of 30 iterations (for more details on the PLS-GAS settings see Ringle et al., 2014); PLS-IRRS settings: $\Delta = 0.0001$, Stop = 30.

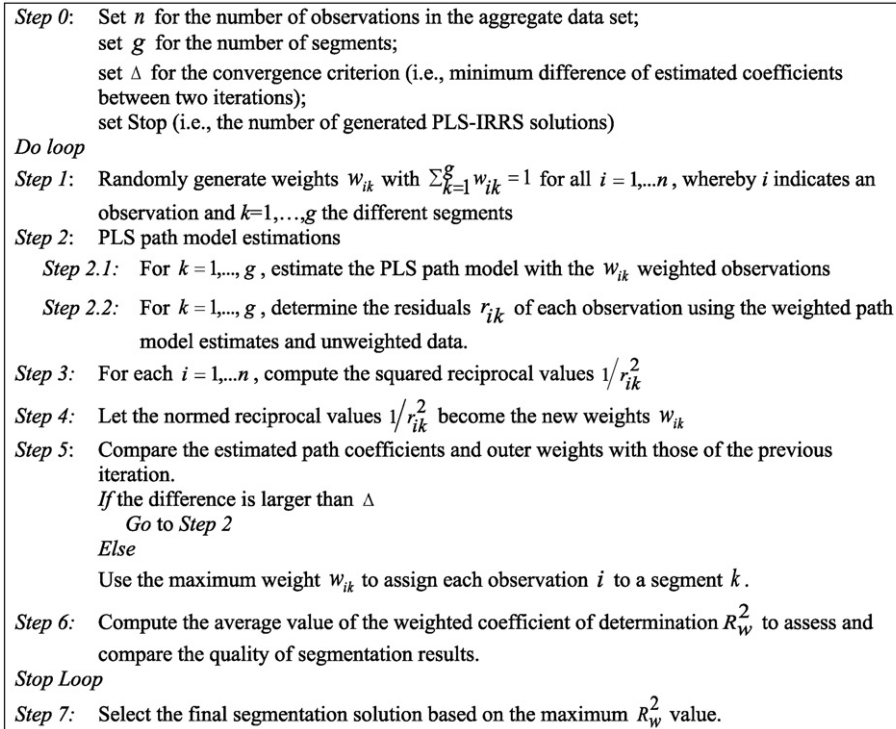


Fig. 1. The PLS-IRRS algorithm.

3. Simulation study to investigate PLS-IRRS's performance

3.1. Design of the study and data generation

The results of a simulation study allow for investigating the behavior and performance of PLS-IRRS. Following Paxton, Curran, Bollen, Kirby, and Chen (2001), the simulation study design accounts for setups commonly encountered in applied research. In line with prior simulation studies on PLS (e.g., Chin, 2003; Henseler & Chin, 2010; Henseler et al., 2014) and particularly Ringle et al.'s (2013, 2014) assessment of PLS-GAS, the design considers the following factors and factor levels:

- Data and segment characteristics*
 - Number of observations for two and three segments [$k = 2$: 50/50, 100/100, 50/150, $k = 3$: 50/50/50].
 - Data distribution of manifest variables [normal, non-normal data with a skewness of 6.183 and an excess of 33.067].
- Measurement model*
 - PLS path model constellations [Mode A; Mode C]. Mode A refers to path models, which only comprise common factor (i.e., reflective) models; Mode C refers to path models which contain both, common factors and composites. All measurement models have four indicators.
 - Pre-defined relationships in the measurement models [high, low, mixed]. Common factors: In the high condition, all outer loadings are at 0.95, whereas in the low condition, all loadings are at 0.75. In the mixed condition, all loadings linearly increase from 0.75 to 0.95. Composites: When the weights are low (high), they have a pre-specified value of 0.20 (0.40). The mixed factor level assigns the low pre-specified weight of 0.20 to the first manifest variable and the high weight of 0.40 to the last manifest variable per composite; in the mixed factor level constellation, weights linearly increase from 0.20 to 0.40.
- Structural model*
 - The error variance of the endogenous construct(s) [5%, 10%, and 20%]. As the path coefficient estimates depend on the pre-specified level of error variance, this research uses Ringle et al.'s (2014) distance

measure D—defined by the weighted averages of the generalized segment-specific coefficients of determination—to control for the separability of the segments.

- Complexity of the PLS path model as shown in Fig. 2 [simple path model, complex path model].

To obtain stable average results, the study generates ten data sets for each factor level combination. For the 288 factor level combinations ($= 4 \cdot 2 \cdot 2 \cdot 3 \cdot 3 \cdot 2$), the simulation study includes the analysis of 2880 data sets and segmentation runs. The data generation follows Ringle et al.'s (2014) procedure, which delivers data sets that precisely match the a priori determined specifications for Mode A and Mode C path model constellations. All computations draw on the GAUSS 9.0 software (Aptech, 2012). While PLS-GAS uses the settings that Ringle et al. (2014) describe in their research (e.g. 100 individuals and 1000 populations in Stage 1; a maximum of 30 iterations in Stage 2), PLS-IRRS draws on ten repetitions and a Δ value of 0.0001 as stop criterion. Finally, supplementing the stand-alone assessment of the method, this research benchmarks PLS-IRRS's performance with that of PLS-GAS, which has been shown to outperform the established FIMIX-PLS and REBUS-PLS procedures (Ringle et al., 2013, 2014).

3.2. Results

To assess and compare the PLS-IRRS and PLS-GAS methods, this research examines the quality of the segmentation solution in terms of correctly reproducing pre-defined data partitions. More precisely, the assessment relies on the hit rates, that is, the proportion of observations generated for a specific segment and finally assigned to that segment.

The analysis results show that, regardless of whether the data are normal or non-normal, PLS-IRRS performs consistently well in all combinations of numbers of observations, numbers of manifest variables in the measurement models, relative segment sizes, and levels of outer loadings. In most constellations, PLS-IRRS achieves hit rates around 80%, that is, the method correctly partitions the data according to the data pre-specification. Compared to PLS-GAS, PLS-IRRS achieves the

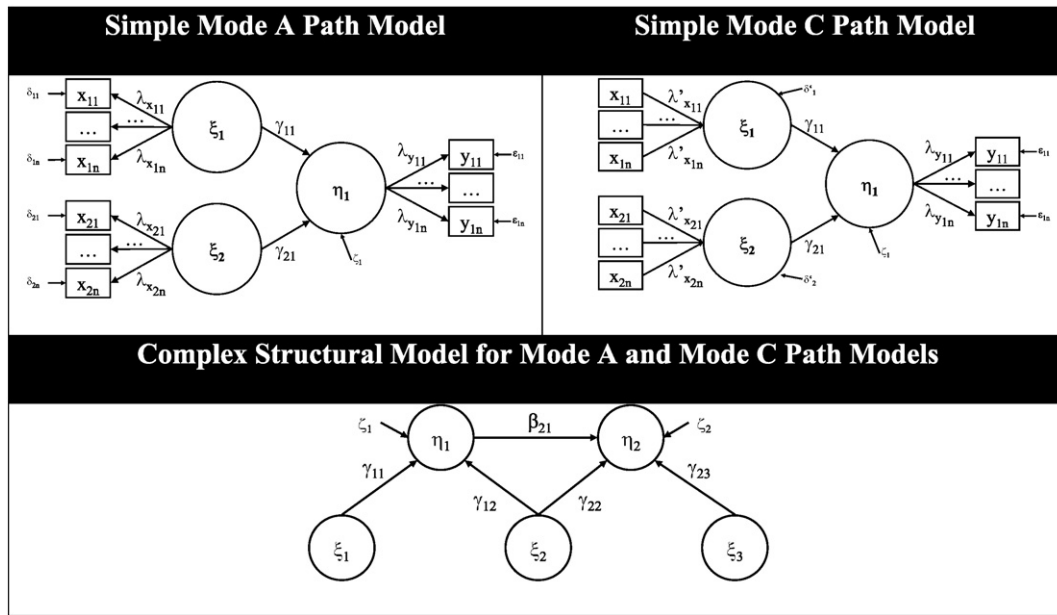


Fig. 2. PLS path models (Ringle et al., 2014). Note: See Ringle et al. (2014) for an explanation of symbols.

same, or slightly higher, hit rates across all factor level constellations for the Mode A path models. This finding also holds for the Mode C path models, however, at slightly lower hit rates of around 65%. For all models, the results also show that (1) higher levels of error variance in the endogenous construct(s) and (2) lower differences in segment-

specific PLS path coefficients, as defined by the distance measure D, entail lower hit rates.

Fig. 3 shows the PLS-IRRS and PLS-GAS hit rates (including 95% confidence bands) of a simple Mode A path model constellation and two segments of normally distributed data with 100 observations

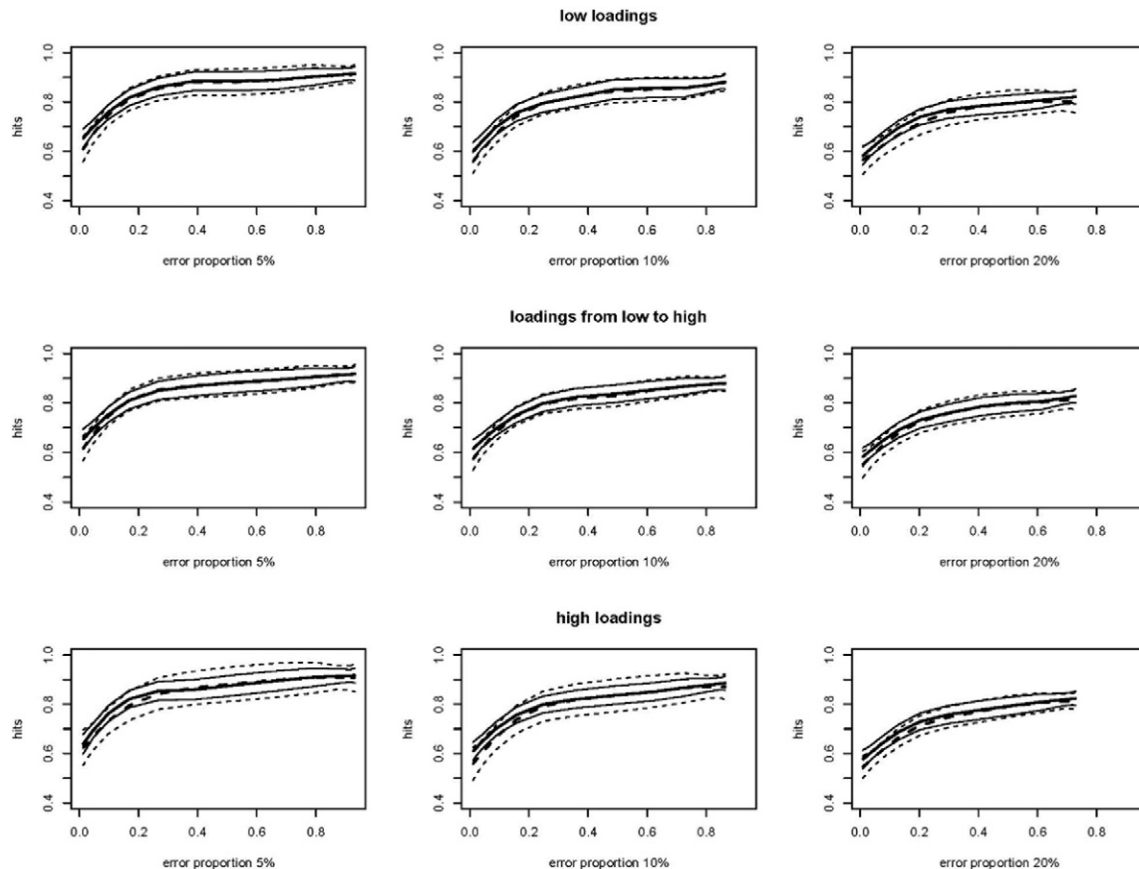


Fig. 3. Hit rates for two segments with 100 observations each (simple Mode A path model and normally distributed data). Notes: x-axis: difference D; y-axis: hits = hit rate; PLS-IRRS results: thick line (95% confidence bands: thin lines); PLS-GAS results: thick dashed line (95% confidence bands: thin dashed lines).

each for (1) varying loading patterns and (2) varying error variances of the endogenous latent variable. The results show that the two methods perform equally well with PLS-IRRS offering slight advantages when segments are not well separated (i.e., for low values in D on the x-axis).

Fig. 4 shows a similar setup but with 50 observations per segment and non-normal data. In these constellations, PLS-IRRS's advantage vis-à-vis PLS-GAS is more pronounced, particularly for high error variances of 20%.

Fig. 5 illustrates the hit rates for a factor constellation consisting of a complex Mode A path model and two segments of normally distributed data with 50 observations each. Again, PLS-IRRS performs at least at the same level of PLS-GAS—this finding analogously holds for all other factor levels.

Most importantly, the run time of PLS-IRRS is much less than that of PLS-GAS across all analyzed situations. Depending on the experimental condition in the simulation study, the PLS-IRRS method is about 50 to 5000 times faster than PLS-GAS (Fig. 6).

To summarize, the results demonstrate the effectiveness and efficiency of PLS-IRRS in handling segmentation tasks. PLS-IRRS performs equal to PLS-GAS—the primary approach in the field (Ringle et al., 2013, 2014)—but at substantially shorter run times. PLS-IRRS's substantial computational speed advantage justifies the establishment of this new segmentation method. More explicitly, the computational speed advantage is so substantial that PLS-IRRS makes PLS-GAS obsolete in most applications.

4. Empirical example

4.1. The ACSI model

The empirical illustration employs the ACSI model (Anderson & Fornell, 2000; Fornell et al., 1996), one of the most prominent PLS

applications, which prior research also used to demonstrate the usefulness of other PLS segmentation methods, such as FIMIX-PLS and PLS-GAS (Rigdon, Ringle, Sarstedt, & Gudergan, 2011; Ringle et al., 2010, 2014). This study uses the same data, model setup, and segment number as these prior studies to ensure comparability of the results.

The authors use SmartPLS 3 (Ringle, Wende, & Becker, 2015) for the estimation of the full data set and the segment-specific models. Consistent with prior studies, the results suggest that the aggregate PLS analysis yields reliable and valid construct measures (for details of the PLS results assessment, see Hair, Hult, Ringle, & Sarstedt, 2017). Table 2 shows that, in the full data set, all paths are significant and positive, but vary in their magnitude from 0.02 (*customer expectation on customer satisfaction*) to 0.69 (*customer satisfaction on customer loyalty*).

4.2. Segmentation

The next step involves running PLS-GAS and PLS-IRRS, whereby the algorithms use the same settings of the previous simulation study. In comparison with the results of the full data set, the R^2_w values are on average 28% and 29% higher when considering two segments (Table 2), underlining the model's superior explanatory power when partitioning the data. To cross-validate the model in terms of out-of-sample prediction—a crucial aspect in model assessment (Gigerenzer & Brighton, 2009; Woodside, 2013)—the analysis further considers the average weighted Stone-Geisser's Q^2_w (Geisser, 1974; Stone, 1974). The computation of this criterion draws on blindfolding, a sample reuse technique, which systematically omits different sets of data points across multiple runs in the endogenous construct's indicators (Chin, 1998; Hair et al., 2017). The results from each blindfolding run allow computing the prediction error of the omitted data points. The (weighted) Q^2 therefore allows for predictive validation testing of

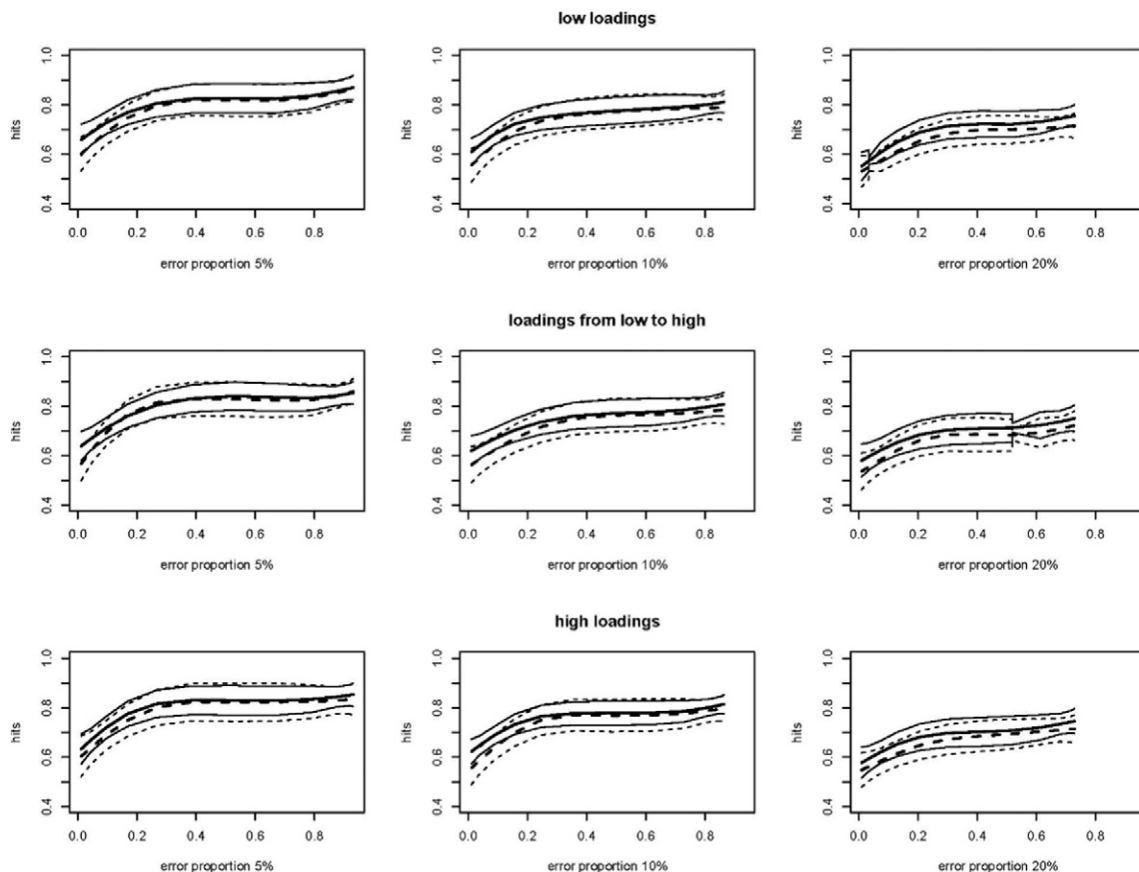


Fig. 4. Hit rates for two segments with 50 observations each (simple Mode A path model and non-normally distributed data). Notes: x-axis: difference D ; y-axis: hits = hit rate; PLS-IRRS results: thick line (95% confidence bands: thin lines); PLS-GAS results: thick dashed line (95% confidence bands: thin dashed lines).

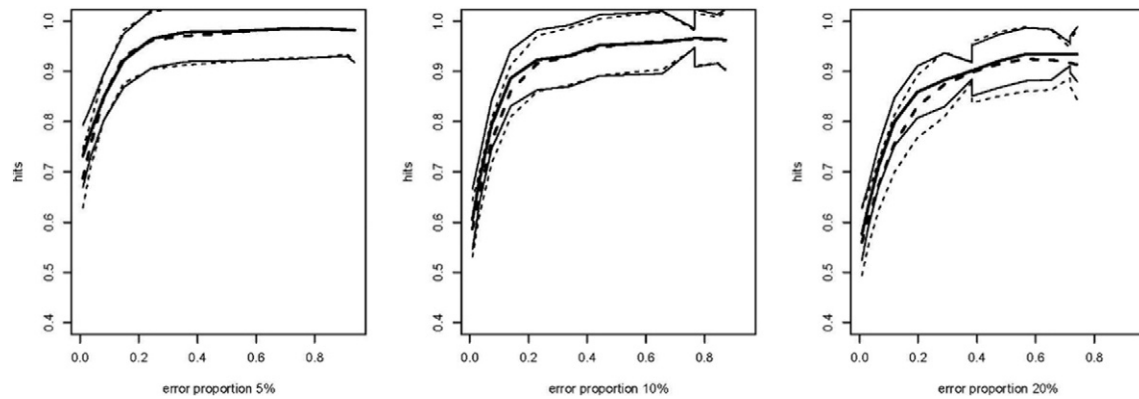


Fig. 5. Hit rates for two segments with 50 observations each (complex Mode A path model, high loadings, and normally distributed data). Notes: x-axis: difference D; y-axis: hits = hit rate; PLS-IRRS results: thick line (95% confidence bands: thin lines); PLS-GAS results: thick dashed line (95% confidence bands: thin dashed lines).

models based on a series of holdout samples, iteratively drawn from the data set. The results in Table 2 show that both the PLS-IRRS and the PLS-GAS show on average an increase in out-of-sample predictive power (Q^2_w) compared with the original aggregated data set. PLS-IRRS shows an increase in predictive performance for *perceived quality* and *perceived value*, while this is not the case for *customer satisfaction* and *customer loyalty*. In contrast, PLS-GAS shows only a substantial increase in Q^2_w for *perceived value*. Interestingly, both procedures uncover one segment with high predictive power and one segment with lower predictive power, whereby this effect is more pronounced for the PLS-IRRS method. This result confirms the tendency observed with regard to the in-sample explanatory power, where PLS-IRRS also produces one segment with particularly high R^2_w values and one segment with lower R^2_w values. Overall, the analysis of predictive validity using the Q^2_w shows the advantageous nature of the segmentation solution, where overfitting seems to be of no particular concern and predictive power is increased.

The segment-specific coefficients of PLS-IRRS are, for most of the paths, significantly different for the two segments of data based on the PLS multigroup analysis (PLS-MGA; Sarstedt, Henseler, & Ringle, 2011b; the Welch-Satterthwait test yield the same results as the PLS-MGA approach). Two exceptions exist: (1) the difference of the paths from *customer expectation* to *customer satisfaction* is not significant, which is not surprising, since each segment-specific path (segment 1: 0.00; segment 2: 0.02) is not significantly different from zero ($p > 0.05$), and (2) the difference of the paths from *perceived quality* to *perceived value* ($\Delta = 0.00$) is not significant. In terms of significant path relationships, the full data model and the segment-specific model estimations show a similar picture (i.e., all path coefficients are significant except the *customer expectation* to *customer satisfaction* relationship). Yet the path coefficient's magnitude changes when considering

the heterogeneity in the model. The path from *customer expectation* to *perceived value*, which is small but significantly positive (0.07, $p < 0.01$) in the full data model, is noteworthy because the path is significantly negative in the first segment (-0.13 , $p < 0.01$) but significantly positive in the second (0.39, $p < 0.01$).

On the surface, PLS-IRRS and PLS-GAS show very similar results. The main difference is that all differences between the segments are significant for PLS-GAS, while this is not the case for two of the differences in PLS-IRRS. Both methods provides similar relative segments' sizes, R^2_w and Q^2_w values, especially for the key target constructs *customer satisfaction* and *customer loyalty*. Further comparing the results of the two segments shows that a specific path coefficient that is larger (smaller) in PLS-IRRS's first segment is also the larger (smaller) path in PLS-GAS. For example, for both, PLS-IRRS and PLS-GAS, the relationship between *customer satisfaction* and *customer loyalty* is weaker in the first than in the second segment. The only exception is the path from *perceived quality* to *perceived value*, where PLS-IRRS does not indicate any difference.

The two methods diverge strongly regarding the amount of time used. In this ACSI model application, PLS-IRRS is more than 1000 times faster than the PLS-GAS procedure (Table 2). On a laptop computer with an Intel Core i7 with 2.7 GHz, 8 GB RAM, and Windows 7, 64 bit, PLS-IRRS needed 41 s, whereas PLS-GAS took 43696 s for the first stage to generate a starting partition, and 21859 s for the second stage (i.e., a total of roughly 18 h). Even a single PLS-GAS run needed, on average, more than one and a half hours for both stages (i.e., 109 min, ranging from 97 to 120). Such a large difference is important for researchers who want to exploit PLS's capabilities as an exploratory research tool (Jöreskog & Wold, 1982) by calculating and comparing different model setups and numbers of segments. In addition, as repeated calculations with more than ten different starting partitions are usually advisable, the run time difference can easily reach days or even weeks. Providing similar results in much less time, means that PLS-IRRS facilitates an effective and efficient assessment of unobserved heterogeneity—as extant research calls for (e.g., Hahn et al., 2002; Hair et al., 2012c).

In summary, both methods provide segmentation solutions with similar patterns of path coefficients and R^2 values, which differ only slightly. However, the final partition of both PLS-IRRS and PLS-GAS shows only a correspondence of 53%, which is not differentiable from a mere chance similarity. The question arises why the two methods provide such different outcomes and which solution to select. As the true segmentation solution is not known and both methods perform equally well in the simulations, this result suggests that at least one approach stops in a local optimum solution. To circumvent this issue, a higher number of segmentation runs (e.g., 100 repetitions) with random starting partitions—instead of carrying out ten runs as done in this example—could lead to better (i.e., more similar) solutions. Since the

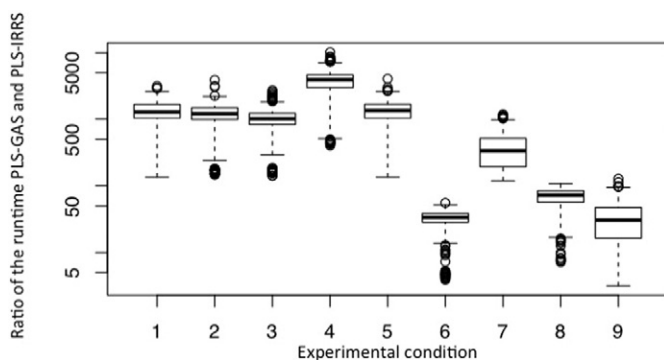


Fig. 6. PLS-GAS run time in relation to PLS-IRRS for nine experimental condition examples from the simulation study.

Table 2
PLS-GAS and PLS-IRRS applications to the ACSI model.

	Full data set	PLS-IRRS			PLS-GAS		
		k = 1	k = 2	Δ ^a	k = 1	k = 2	Δ ^a
Segment size	100%	49.75%	50.25%		49.02%	50.98%	
	10418	5183	5235		5107	5311	
Customer satisfaction → customer loyalty	0.69*	0.43*	0.84*	0.41*	0.64*	0.74*	0.10*
Customer expectation → customer satisfaction	0.02*	0.00	0.02	0.02	−0.01	0.05*	0.06*
Customer expectation → perceived quality	0.56*	0.37*	0.81*	0.44*	0.53*	0.64*	0.11*
Customer expectation → perceived value	0.07*	−0.13*	0.39*	0.52*	−0.08*	0.29*	0.37*
Perceived quality → customer satisfaction	0.56*	0.61*	0.54*	0.07*	0.64*	0.48*	0.16*
Perceived quality → perceived value	0.62*	0.49*	0.49*	0.00	0.67*	0.55*	0.12*
Perceived value → customer satisfaction	0.39*	0.35*	0.41*	0.06*	0.32*	0.45*	0.13*
	Full data set	k = 1	k = 2	Weighted average	k = 1	k = 2	Weighted average
R ² customer satisfaction	0.78	0.69	0.85	0.77	0.76	0.80	0.78
R ² customer loyalty	0.47	0.19	0.71	0.45	0.41	0.54	0.48
R ² perceived quality	0.31	0.13	0.66	0.40	0.28	0.41	0.35
R ² perceived value	0.44	0.21	0.71	0.46	0.40	0.58	0.49
Mean R ²	0.50	–	–	0.52	–	–	0.52
Q ² customer satisfaction	0.63	0.50	0.73	0.61	0.61	0.65	0.63
Q ² customer loyalty	0.47	0.19	0.71	0.45	0.41	0.54	0.48
Q ² perceived quality	0.23	0.08	0.55	0.31	0.22	0.27	0.25
Q ² perceived value	0.39	0.17	0.66	0.42	0.34	0.52	0.43
Mean Q ²	0.43	–	–	0.45			0.45

^a PLS multigroup analysis: PLS-MGA and Welch-Satterthwait test; both approaches yield the same results in terms of significance of the differences (Sarstedt et al., 2011b).

* Significant at 0.01.

run time of PLS-GAS is extremely high, this study exercises the 100 runs with random starting partitions only for PLS-IRRS. The best solution of these 100 runs is identical to the PLS-IRRS solution that ten runs provide. This finding gives some additional confidence that PLS-IRRS does not converge in a local optimum.

4.3. Qualitative comparative analysis using fuzzy sets

To ensure the validity of results based on regression-type analyses, researchers must ensure that they did not tap into a situation of complex causality (Schneider & Eggert, 2014). The fuzzy set qualitative comparative analysis (fs/QCA; Ragin, 2008a, 2009) allows capturing complex causality in the presence of causal asymmetry (Fiss, 2011; Wagemann, Buche, & Siewert, 2016; Woodside, 2013) in empirical studies (e.g., Woodside, Hsu, & Marshall, 2011; Woodside, Ko, & Huan, 2012). More specifically, this study uses the full data set and segment-specific data sets from the PLS-IRRS segmentation to explore the causal mechanisms that lead to high (low) customer satisfaction in the ACSI model. The analysis draws on the fs/QCA 2.5 software package (Ragin & Davey, 2014).

To begin with, the analysis requires extracting the unstandardized latent variable scores from the PLS analysis and their transformation into fuzzy set membership scores. The 75th percentile of each variable serves as an anchor for the full membership, the 50th percentile as the crossover point, and the 25th percentile as an anchor for full non-membership (Fiss, 2011). Computation of these transformations uses the continuous calibration procedure of the fs/QCA 2.5 software.

The next step involves setting the frequency and consistency thresholds of the truth-table analysis. In the three data sets (i.e., the aggregate level data and the two segment-specific data sets), all possible causal set conditions have at least 100 observations, which is far above the minimum frequency recommendations of, for example, one and ten (Ragin, 2008b; Woodside, 2013). Hence, this study includes all possible causal set conditions in the truth-table analyses of all the data sets, which avoids problems of limited diversity. For the consistency threshold, the minimum recommended value is 0.75 (Fiss, 2011; Ragin, 2006, 2008a). Consistency values below this minimum threshold indicate the presence of substantial inconsistency. Ideally, consistency scores exceed a value of 0.90 (Legewie, 2013). Since cutoff points are to some extent

arbitrary, researchers often choose a cutoff point based on ordering the consistency values in the truth-table analysis and by choosing a value that shows a larger gap to the next lower consistency value (Wagemann et al., 2016). In this study is a relatively large gap between consistency values above 0.89 and those clustered around 0.74 to 0.82. Following the recommendations and using the higher cutoff value of 0.89 still entails relatively high coverage values in this study (Table 3), which are clearly above the critical value of 0.5 (Legewie, 2013).

When analyzing the presence and absence of the focal outcome *customer satisfaction* (Table 3), the fs/QCA 2.5 results provide support for causal symmetry. In both cases (i.e., presence and absence of the focal outcome *customer satisfaction*), the same configuration emerges in the overall data set as well as in each segment. This configuration comprises the presence (absence) of both *perceived quality* and *perceived value* as the only core configuration in the complex, parsimonious and intermediate solution that leads to presence (absence) of *customer satisfaction*. Hence, the results show no sign of causal equifinality or asymmetry (Fiss, 2011; Wagemann et al., 2016), while providing a very good fit in terms of high consistency (≥ 0.89) and coverage (≥ 0.55). These findings substantiate a correlation-based symmetric analysis of causality when examining and interpreting the PLS path coefficient estimates.

5. Discussion and future research

This research introduces PLS-IRRS, a new latent class approach to PLS, which allows for uncovering unobserved heterogeneity in path model structures. Building on the concept of M-estimators (Schlittgen, 2011), the method identifies homogeneous segments of observations based on the endogenous constructs' residual values. The PLS-IRRS method does not rely on distributional assumptions and uncovers heterogeneity in both common factor and composite models. Thereby, the approach is generally applicable to segmentation tasks in PLS path models commonly encountered in business research studies. Results from a large-scale simulation study show that across a wide range of data constellations, the PLS-IRRS results closely match the expectations from the artificial data sets. In comparison to PLS-GAS, the hitherto best PLS-based latent class approach (Ringle et al., 2013, 2014), PLS-IRRS performs equally well. Most importantly, in contrast to PLS-GAS, PLS-IRRS computations are multiple times faster, making the method

Table 3
fs/QCA 2.5 results.

	Presence			Absence		
		Coverage	Consistency		Coverage	Consistency
Full data set	Perceived quality * Perceived value	0.67	0.91	~Perceived quality * ~ Perceived value	0.69	0.91
$k = 1$	Perceived quality * Perceived value	0.55	0.89	~Perceived quality * ~ Perceived value	0.62	0.89
$k = 2$	Perceived quality * Perceived value	0.82	0.92	~Perceived quality * ~ Perceived value	0.77	0.94

Note: * indicates a logical AND; ~ indicates a logical negate; unique and raw coverage are equivalent because only one configuration exists.

particularly advantageous for PLS applications, which often entail estimating multiple models in a knowledge discovery process. As [Wold \(1982, p. 29\)](#), the originator of the PLS method note, “the investigator uses the first design to squeeze fresh knowledge from the data, thereby putting flesh on the bones of the model, getting indications for modifications and improvements, and gradually consolidating the design.” However, such an endeavor requires estimating different model configurations in a reasonable amount of time. PLS-GAS does not allow for such an assessment ([Matthews et al., 2016](#)). The combination of its general applicability, its high quality of the results, and the computational speed justifies the conclusion that PLS-IRRS is superior, in comparison to not only PLS-GAS but also FIMIX-PLS and REBUS-PLS. Therefore, PLS-IRRS should be the primary method of choice when assessing whether unobserved heterogeneity critically affects PLS results ([Becker et al., 2013b](#)).¹

Despite the apparent usefulness of PLS-IRRS, the approach and this study have certain limitations, which offer at least six opportunities for further research.

- (1) Like other latent class methods (e.g., PLS-GAS, FIMIX-PLS or REBUS-PLS), PLS-IRRS requires the researcher to pre-specify the number of segments to retain from the data, which is typically unknown. The systematic selection of the most appropriate number of segments involves running PLS-IRRS for different segment numbers. To guide the selection process, researchers can use the weighted average of segment-specific coefficients of determination R_w^2 ([Schlittgen, 2011](#)) and choose the solution, which yields the highest weighted average R_w^2 , thereby indicating the solution with the greatest in-sample explanatory power. In addition, [Schlittgen \(2011\)](#) describes a bootstrap-based test for clusterwise regressions, which is also adaptable to PLS, to test the significance of the difference in the weighted average R_w^2 for alternative numbers of segments. The basic idea is to test the null hypothesis that $R_{w,g+1}^2 - R_{w,g}^2 = 0$ against the alternative hypothesis that $R_{w,g+1}^2 - R_{w,g}^2 > 0$, whereby g indicates a certain number of (pre-specified) segments. This test is similar to a bootstrap-based test of the likelihood ratio in mixture models ([McLachlan, 1987](#)). However, focusing on the average weighted R_w^2 may entail problems. First, the average weighted R_w^2 does not penalize model complexity. By contrast, segment retention criteria in mixture models account for the model complexity by adding a penalty to the likelihood (e.g., AIC or BIC). Since the average weighted R_w^2 does not have such a feature, the criterion may point towards selecting a segmentation solution with too many segments. The bootstrap-based significance test of differences may solve this problem but needs more research in a PLS context. Second, relying on purely data driven assessment may result in segmentation solutions that are neither theoretically nor practically relevant. Researchers should not only focus on statistical criteria, but they should also consider model parsimony and practical relevance of the segmentation solution. [Becker et al. \(2013b\)](#) propose an unobserved

heterogeneity discovery process, which aims at guiding researchers to discover the most suitable number of segments while maintaining practical and theoretical relevance of the findings. For example, researchers should not consider an additional segment that increases the average weighted R_w^2 , when the segment is too small to obtain practical relevance or the segment is not differentiable from other segments (i.e., not statistically different). Concerning the latter aspect one should note that—with increasing sample sizes—even small differences could become significant and, thus, the difference between segments should be substantial as well. Finally, the segmentation solutions should be explainable by external variables to turn unobserved heterogeneity into observed heterogeneity, which is testable in future studies.

- (2) Clusterwise regression and PLS segmentation problems are difficult to solve due to the target function's possibly large number of local optimum solutions. In line with other PLS segmentation methods such as FIMIX-PLS and PLS-GAS, PLS-IRRS requires several runs with random starting partitions per pre-defined number of segments to ensure that the final solution is as close as possible to the global optimum. Related research on FIMIX-PLS ([Sarstedt, Becker, Ringle, & Schwaiger, 2011a](#)) finds that researchers should use at least ten repetitions to ensure that the result is close to the optimum segmentation solution. This recommendation is of course not directly transferable to PLS-IRRS because the mixture modeling's expectation–maximization algorithm and the PLS-IRRS approach are different in nature. Generally, the number of repetitions to arrive at solutions that are close to the global optimum depends on the unknown separability of the empirical segmentation task. No algorithm can ensure a reliable solution in such segmentation tasks that imply a discrete optimization problem. Researchers always need to conduct sufficient repetitions to explore the solution space to avoid local optima. Future research should investigate the number of repetitions necessary to identify a segmentation solution that is at least close to the global optimum in PLS-IRRS. While this study uses ten repetitions, which prior research advocates for FIMIX-PLS ([Sarstedt et al., 2011a](#)), an independent assessment of this issue is useful to guide practical applications.
- (3) This studies' simulation design only includes path models with purely common factors and path models with mixed common factors and composites. The study does not include purely composite models. [Becker, Rai, and Rigdon \(2013a\)](#) propose a method to simulate data from covariance matrices that define composite-based population models. Analyzing purely composite models is another promising area for future research. PLS-IRRS does not build on the type of measurement in its algorithm but focuses on the residuals of structural relations between latent variables. Hence, the authors expect a similarly high performance for purely composite models as in the mixed (common factor-composite) model.
- (4) Aligning PLS-IRRS with the new consistent PLS method (PLSc; [Dijkstra, 2014; Dijkstra & Henseler, 2015](#)) is another important advancement in future research. PLSc extends the basic PLS method to common factor models by disattenuating the correlations between pairs of latent variables. In light of the

¹ PLS-IRRS is available in the R package SEGIRLS_0.5 at http://www.wiso.uni-hamburg.de/fileadmin/bwl/statistikundoeconometrie/Schlittgen/SoftwareUndDaten/SEGIRLS_0.5.tar.gz

considerable potential that PLSc holds for future research (Henseler, Hubona, & Ray, 2016b; Rigdon, 2014; Sarstedt, Ringle, Henseler, & Hair, 2014b), extensions of PLS-IRRS algorithm should also draw on PLSc results. A simple approach would be to apply PLS-IRRS in the first stage of PLSc, which uses the basic PLS path modeling results as basis for disattenuating the correlations between pairs of latent variables. Hence, one could use PLSc on the segment-specific path models. However, such an approach might not provide its full potential as the method would use attenuated path coefficients for the optimization of the segments-specific models (i.e., for the calculation of residuals). Supposedly, the better approach is to calculate the residuals in the reweighting process based on disattenuated path coefficients. Yet consistency implies unbiasedness at large samples, while segmentation always implies a reduction of sample size for the segment-specific models. Thus, the advantage of PLSc might not be of particular importance in a segmentation study, which involves small samples at the segment level. Nevertheless, given the importance of the topic, future research should shed more light on the interplay between PLS-IRRS and PLSc.

- (5) Future research may also investigate the combination of uncovering unobserved heterogeneity and testing for causal asymmetry. Discovery of unobserved heterogeneity in the path coefficients is itself already a form of discovery of causal asymmetry. The causal mechanism is not the same for all segments. Certain constructs may not entail a causal influence in all segments or may entail a causal influence with a reversed sign (positive/negative). Yet the segment-specific results are still regression estimates and therefore represent symmetric hypotheses. However, the causal mechanism can even vary to a stronger degree across segments. For example, different configurations may emerge across segments and one segment could involve an antecedent in a necessary causal relationship while another segment involves the same antecedent in a sufficient causal relationship. Existing methods for uncovering unobserved heterogeneity in PLS (as well as regression-type analyses) might not be fully suited to discover segments that vary beyond the direction of regression coefficients. Future research should therefore address this aspect and investigate the ability of methods to uncover heterogeneity in complex causal mechanisms and asymmetric causal relationships.
- (6) Finally, in applying PLS-IRRS, researchers should keep in mind that predictive validation testing of models using holdout samples or cross-validations using blindfolding is a crucial step in the validation process (Gigerenzer & Brighton, 2009; Woodside, 2013). As Woodside (2013), p. 466 notes, “the critical question is whether or not a model (e.g., an empirical multiple regression model or an algorithm) predicts a dependent variable in additional samples—holdout samples that are separate data sets from the data sets used to test the fit of data to a theory.” Good model fit does not necessarily mean that researchers found a good model that is generalizable beyond the sample used for estimation and poor predictions usually result from the best fitting model (Gigerenzer & Brighton, 2009). This problem also poses a threat to segmentation solutions that may fit the sample data very well (e.g., in terms of the explained variance per segment), but perform poorly on other samples that have not been used for estimation. Overfitting the data by forming too many segments, for example, is a common problem of segmentation applications (Andrews, Brusco, Currim, & Davis, 2010a). Engaging in “fishing expeditions” where researchers try to improve the model fit (or significance of certain parameters) by using a segmentation algorithm without theoretical underpinnings is another problem (Becker et al., 2013b). Thus, researchers should routinely assess the

predictive validity of their segmentation solution by using hold-out samples or cross-validation techniques (like the blindfolding procedure) to critically examine whether the segmentation solution generalizes to other samples. In a latent class analysis, hold-out sample validation is particularly challenging as the segmentation solution is—by definition—latent. To allow for a comparison of model estimates with those from a holdout sample, researchers must identify one or more explanatory variable(s) that match the latent class method's partition in the best possible way. Researchers can then partition the original and holdout sample data along the lines of the explanatory variables and compare the segmentation solutions using multigroup analysis (e.g., Sarstedt et al., 2011b). However, finding appropriate explanatory variables to guide this partitioning process is an obstacle that is hard to overcome in empirical research (e.g., Rigdon, Ringle, & Sarstedt, 2010; Sarstedt, Ringle, & Gudergan, 2016). Sample reuse techniques like blindfolding provide a solution to this problem although they might be inferior in assessing predictive validity of the results (e.g., Shmueli, Ray, Velasquez Estrada, & Chatla, 2016). When researchers avoid the overfitting problem, segmentation offers a means to improve the predictive abilities of a model (Andrews, Brusco, & Currim, 2010b; Currim, 1981) and to safeguard the validity of results (Becker et al., 2013b; Jedidi et al., 1997).

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